

Technical Reference Manual

Automated Security & Deterrent System (ESP32-CAM)

Hardware Version: Rev 3.0

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Chapter 1

Executive Summary & System Overview

The Automated Security & Deterrent System is a localized, standalone security endpoint designed to detect human intrusion, capture photographic evidence, and deploy a non-lethal deterrent (pepper spray) automatically.

The core engineering challenge addressed in Hardware Rev 3.0 is aggressive power optimization. Because the system relies on high-draw components (an ESP32-CAM, a 24GHz mmWave radar, and a high-torque MG996R servo), continuous operation would quickly deplete a standard battery supply.

To solve this, the architecture implements a **Deterministic Multi-Sensor Fusion** approach combined with **Dual-Rail Power Gating**. Peripheral sensors and actuators are physically isolated from the power rails during IDLE states, resulting in near-zero quiescent current leakage from the peripherals.

Chapter 2

Power Management & Distribution Architecture

The power delivery network (PDN) is divided into two distinct, switchable rails (3.3V and 5V), both dictated by a master control signal originating from the central microcontroller.

2.1 Input Power & Transient Protection

The board accepts power via a standard 2.1mm DC Barrel Jack or direct wire-to-board leads. A three-stage protection circuit ensures survivability:

Overcurrent Protection (F1)

A fast-acting surface-mount fuse (047603.5MR, rated 3A/4A) serves as the primary fail-safe against catastrophic short circuits, especially considering the high stall current of the servo motor.

Reverse Polarity Protection (D2)

An SS34 Schottky Barrier Rectifier is placed in reverse bias across the input. In the event of inverted power connection, D2 intentionally shorts the circuit, safely blowing F1 before reverse voltage can reach the ESP32 or regulators.

Surge & Transient Clamping (D1)

An SMAJ5.0A TVS diode clamps high-voltage transients and inductive spikes (often caused by servo deceleration) to a safe threshold.

2.2 Dual-Rail Power Gating Subsystem

The defining feature of this PCB is its ability to enter a “Deep Sleep” state where only the ESP32 remains powered.

The ESP32 asserts a HIGH logic level on the CTRL_SIGNAL line to wake peripheral rails.

2.3 3.3V Regulation (Buck Converter)

The 3.3V logic rail powers the HLK-LD2420 radar and the LDR network.

- **Regulator:** TPS563200 synchronous step-down converter
- **Output Current:** 3A
- **Switching Frequency:** 650kHz
- **Enable Control:** CTRL_SIGNAL tied to EN pin

When CTRL_SIGNAL = LOW, the buck converter enters shutdown mode and reduces internal draw to less than $10\mu\text{A}$.

2.4 5V Load Switching (High-Side Switch)

The 5V rail powers:

- HC-SR505 PIR Sensor
- High Power LEDs
- MG996R Servo

Hardware Implementation

- Q3: BSP75N / 2N7002 (Signal Switch)
- Q4: AO3401A (Primary P-Channel MOSFET)

Operating States

ON State: CTRL_SIGNAL HIGH $\rightarrow$ Q3 ON $\rightarrow$ Pulls Q4 Gate LOW $\rightarrow$ Q4 conducts $\rightarrow$ 5V rail active.

OFF State: CTRL_SIGNAL LOW $\rightarrow$ Q3 OFF $\rightarrow$ Pull-up resistor drives Q4 Gate HIGH $\rightarrow$ 5V rail disabled.

Chapter 3

Microcontroller & Control Logic

3.1 Central Processor Unit

The system is governed by the ESP32-CAM module, responsible for:

- Localized decision-making
- WiFi mesh communication
- SD card image logging
- Telegram API-based remote alerting

3.2 Hardware Errata & Patch Notes (Rev 3.0)

Initial schematic issues involved GPIO 0 being used for CTRL_SIGNAL, causing boot conflicts due to its bootstrap function.

Applied Hardware Patch

1. Trace cut between GPIO 0 and CTRL_SIGNAL
2. Isolation of R13 pull-up from 3.3V rail
3. Jumper wire routing GPIO 12 → CTRL_SIGNAL
4. Multiplexing LDR onto GPIO 4

This modification eliminated brownout resets and startup oscillations.

3.3 Final GPIO Pinout Mapping

GPIO Pin	Function Name	Direction	Hardware Description
GPIO 12	CTRL_SIGNAL	OUTPUT	Master Power Switch. HIGH = Buck EN / PMOS Gate ON
GPIO 1	LED_TRIG	OUTPUT	Drives Q1/Q2 (BSP52T1G) for High Power LEDs
GPIO 3	RADAR_RX	INPUT	UART data from HLK-LD2420
GPIO 4	SERVO_LDR_MUXN/OUT		Analog LDR read (IDLE) / PWM Servo (ACTION)
GPIO 13	PIR_TRIG	INPUT	Digital HIGH from HC-SR505
GPIO 16	Auxiliary	N/A	Routed to expansion connector J5 (Pin 2)

Chapter 4

Sensor Fusion Subsystem

To eliminate false-positive triggers (wind, shadows, small animals), the board implements concurrent multi-domain verification.

4.1 mmWave Radar Detection (HLK-LD2420)

- Operating Voltage: 3.3V (Switched Rail)
- Communication: UART over GPIO 3
- Technology: 24GHz FMCW Radar

The radar detects micro-movements, including respiration, and serves as the primary high-sensitivity trigger.

4.2 Passive Infrared Detection (HC-SR505)

- Operating Voltage: 5V (Switched Rail)
- Output: Digital HIGH to GPIO 13

The system confirms intrusion only when:

$$\text{Radar} = \text{TRUE} \wedge \text{PIR} = \text{TRUE}$$

4.3 Ambient Light Sensing (GL5516 LDR)

- Operating Voltage: 3.3V
- Configuration: Voltage divider with 100k Ω resistor
- Interface: GPIO 4 (Multiplexed)

Determines Night Mode operation and controls LED flash activation.

Chapter 5

Actuator & Output Subsystems

The output stage of the system is strictly isolated to the 5V switched rail to prevent inductive feedback and high-current voltage droop from affecting the ESP32 stability.

5.1 Deterrent Deployment (MG996R Servo)

- **Operating Voltage:** 5V (Switched Rail)
- **Control Pin:** GPIO 4 (PWM)
- **PWM Frequency:** 50Hz

The MG996R is a high-torque mechanical servo responsible for actuating the pepper spray canister.

Calibration Specifications

- Neutral/Closed (90°): $1500\mu s$ pulse width
- Actuation Sweep Range: $800\mu s - 2200\mu s$

The calibrated sweep ensures full mechanical depression of the spray nozzle while preventing gear overstrain.

5.2 Visual Deterrent & Camera Flash (High Power LEDs)

- **Operating Voltage:** 5V (Switched Rail)
- **Control Pin:** GPIO 1

Two parallel high-power LED arrays provide:

- Illumination for low-light image capture
- Temporary visual disorientation deterrence

Because GPIO 1 cannot source sufficient current, it drives the base of a high-gain switching stage:

- BSP52T1G (NPN Darlington), or
- BSP75N (N-Channel MOSFET)

This configuration isolates the microcontroller from LED surge currents.

Chapter 6

Operational Algorithm & State Machine

The firmware utilizes a non-blocking deterministic state machine to manage:

- Power sequencing
- Sensor verification
- Servo deployment
- Mesh-network communication

6.1 Phase 1: Boot & Stabilization

1. **Hard Reset:** ESP32 asserts CTRL_SIGNAL LOW.
2. **Brownout Prevention Delay:** 2000 *ms* stabilization delay.
3. **Peripheral Power-Up:** CTRL_SIGNAL HIGH. Sensors initialize for 1000 *ms*.

6.2 Phase 2: IDLE & Scan

1. HLK-LD2420 streams continuous distance data.
2. If target distance < 70 *cm*, transition to Detection State.

6.3 Phase 3: Detection & Verification

1. Sensors remain active for exactly 10 seconds.
2. PIR sensor polled continuously.

$$\text{Trigger Action} = (\text{Radar} = \text{TRUE}) \wedge (\text{PIR} = \text{TRUE})$$

6.4 Phase 4: ACTION (Spray Sequence)

1. **Sensor Blackout:** CTRL_SIGNAL LOW
2. **Deploy:** Servo actuates programmed spray cycles
3. **Restore:** Sensors re-enabled
4. **Capture & Mesh:**
 - Image captured to SD Card
 - Telegram API transmission
 - ESP-NOW mesh broadcast

6.5 Phase 5: Cooldown

A fixed 20-second lockout prevents repeated deployment before returning to IDLE.

Chapter 7

Design Calculations & Empirical Measurements

7.1 TPS563200 Buck Converter Feedback Calculation

The output voltage is determined by:

$$V_{OUT} = 0.765 \times \left(1 + \frac{R14}{R15}\right)$$

Using:

- $R14 = 33k\Omega$
- $R15 = 10k\Omega$

$$V_{OUT} = 0.765 \times \left(1 + \frac{33000}{10000}\right)$$

$$V_{OUT} = 0.765 \times 4.3 = 3.2895V$$

Result: Output $\approx 3.3V$, within ESP32 and radar tolerances.

7.2 Power Consumption Profiles

System State	3.3V Rail	5V Rail	System Total
Deep Sleep	$\approx 0mA$	$\approx 0mA$	$< 5mA$
IDLE / Scan	$\approx 80mA$	$\approx 2mA$	$\approx 280mA$
ACTION	$0mA$ (Gated)	$1.0A-2.5A$ Peak	$> 2.0A$

These measurements validate the effectiveness of the Dual-Rail Power Gating architecture.

Chapter 8

Bill of Materials (BOM) & Key Components

Designator	Component	Value / Number	Part	Description
U1	Buck Converter	TPS563200		4.5V–17V Input, 3A Sync Step-Down
Q4	P-MOSFET	AO3401A		30V, 4A Logic-Level PMOS
Q3	N-Switch	2N7002 / BSP75N		Gate Control Signal Switch
Q1, Q2	LED Driver	BSP52T1G BSP75N	/	LED High Current Driver
F1	Fuse	047603.5MR		3.5A Fast Acting SMD Fuse
D1	TVS	SMAJ5.0A		5V Transient Suppressor
D2	Diode	SS34		Schottky Reverse Protection
C4	Bulk Capacitor	470μF / 16V		Input Filtering
C3	Servo Capacitor	470μF / 16V		Servo Surge Buffer
C7, C9	Decoupling	100nF		Ceramic Noise Filter

Chapter 9

Hardware Schematics & PCB Layout

9.1 Schematic Diagram

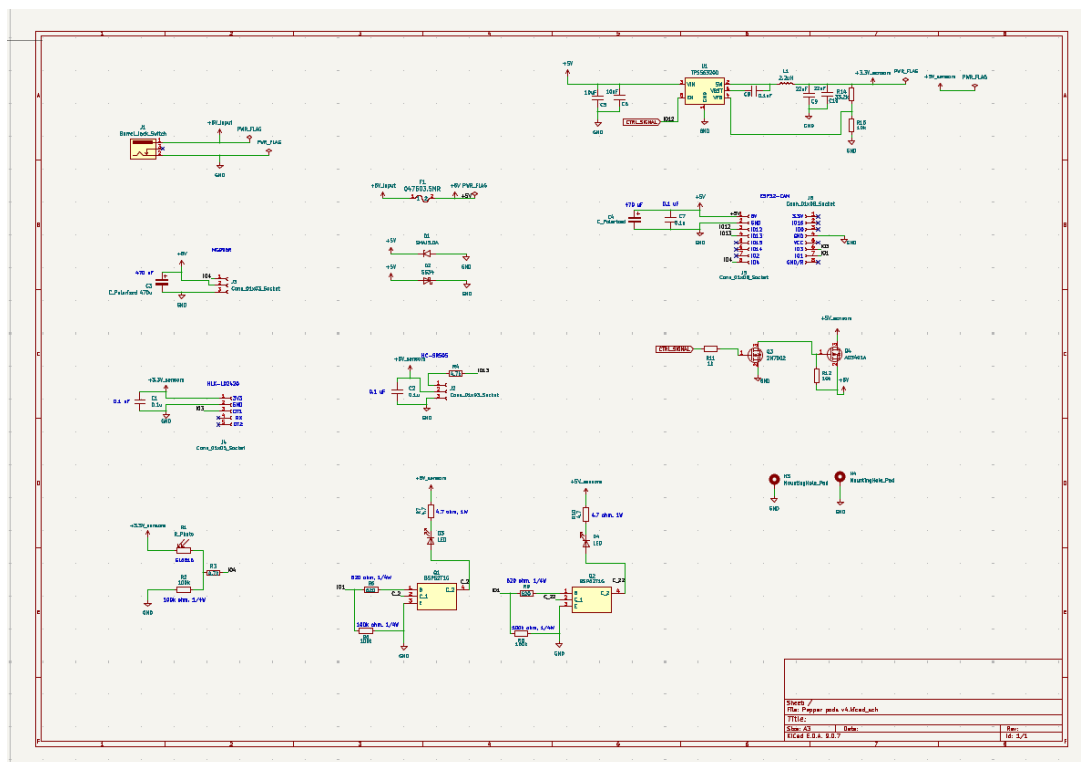


Figure 9.1: Rev 3.0 Schematic Diagram Showing Updated GPIO 12 Control Logic

9.2 PCB Layout (Top Layer)

9.3 PCB Layout (Bottom Layer)

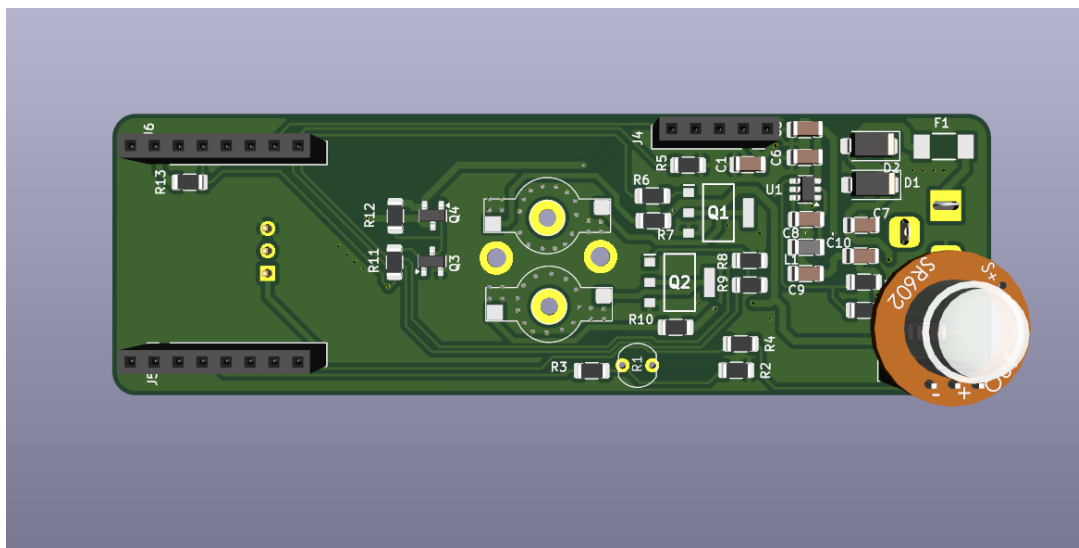


Figure 9.2: Top Copper Layer with High-Current 5V Traces Highlighted

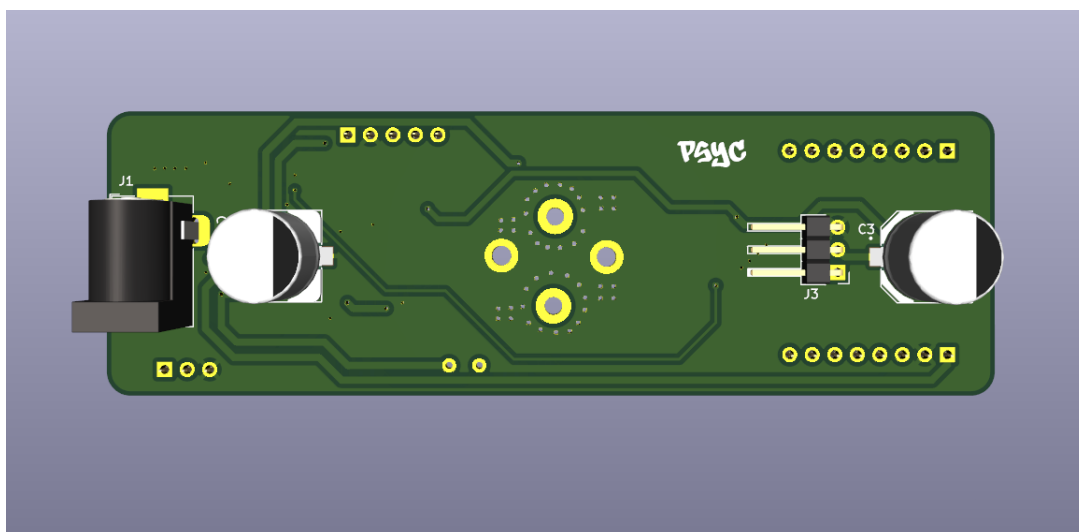


Figure 9.3: Bottom Layer Showing Continuous Ground Plane and Thermal Vias

Chapter 10

Firmware & Open Source Repository

The complete firmware codebase, including:

- ESP-NOW Mesh Networking
- Telegram API integration
- SD Card buffering

is available at:

- **GitHub Repository:** <https://github.com/noeljoyce6/Pepper-Spray-Board>
- **Framework:** Arduino Core for ESP32
- **Required Libraries:**
 - ESP32Servo
 - ArduinoJson
 - esp_camera

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